

Malicious Domains Analysis Report 2026

VirusTotal Multi-Engine Platform · 162 Domain Records

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§1 Abstract

This report presents a comprehensive analysis of the Malicious Domains dataset comprising 162 domain records collected via VirusTotal's multi-engine threat intelligence platform during 2026. The analysis focuses on predicting threat severity (Low, Medium, High) using threat vote counts, reputation metrics, and domain structural features.

Five regression models were evaluated. The Random Forest Regressor demonstrated superior practical performance with an R^2 score of 0.839 and RMSE of 0.141. The Malicious_Votes feature emerged as the dominant severity predictor (48.3% importance), with crowd-sourced voting metrics significantly outperforming domain structural features (length, TLD, hyphen presence).

§2 Introduction

Malicious domain infrastructure serves as entry points for phishing campaigns, malware distribution platforms, command-and-control (C2) servers, and information exfiltration conduits. VirusTotal's platform aggregates security engine verdicts from 91 independent malware analysis engines, enabling correlative threat assessment through voting consensus.

Analytical objectives:

- Characterize threat severity distribution across 162 malicious domain records
- Identify domain structural features predictive of threat severity
- Develop regression models predicting Threat_Severity from domain characteristics
- Evaluate model performance through cross-validation
- Establish domain triage prioritization matrices for security operations

§3 Dataset Description

3.1 Threat Severity Distribution

Severity	Count	Percentage	Description
Low	145	89.5%	Minimal threat signals; low confidence malicious classification
Medium	10	6.2%	Moderate threat signals; mixed engine verdicts
High	7	4.3%	Strong threat signals; high confidence malicious classification
TOTAL	162	100.0%	Complete dataset

Class Imbalance Challenge: The 89.5% Low severity concentration creates significant modeling challenge, with minority classes (Medium + High) representing only 10.5% of training data. SMOTE oversampling is recommended for production deployment.

3.2 Top-Level Domain (TLD) Distribution

TLD	Count	Percentage	Registrar Concentration
.com	59	36.4%	GoDaddy, Name.com, Namecheap
.net	33	20.4%	Network Infrastructure TLD
.org	28	17.3%	Non-profit/Organization TLD
.io	9	5.6%	Technology/Startup TLD
.ch	8	4.9%	Switzerland country code
.ru	7	4.3%	Russia country code
.xyz	5	3.1%	Generic TLD
.info	4	2.5%	Information TLD
Others	6	3.7%	Miscellaneous TLDs

3.3 Domain Length Distribution

Length Range	Count	Percentage	Characteristics
5–9 chars	18	11.1%	Highly memorable; premium registrations
10–14 chars	41	25.3%	Moderate memorability; common size
15–19 chars	52	32.1%	DOMINANT (one-third of dataset)
20–24 chars	31	19.1%	Longer, harder to remember
25+ chars	20	12.3%	Very long; often typosquatting domains

Mean domain length: 15.5 characters ($\sigma = 5.2$ characters).

3.4 Voting Metrics Distribution

Metric	Mean	Std Dev	Min	Max
Malicious_Votes	1.31	3.34	0	17
Harmless_Votes	38.9	23.1	0	78
Suspicious_Votes	0.31	1.12	0	8
Undetected_Votes	50.3	24.7	8	91

3.5 Correlation Analysis

Feature Pair	Correlation	Interpretation
Malicious_Votes ↔ Threat_Severity	+0.892	VERY STRONG positive correlation
Reputation ↔ Threat_Severity	−0.678	Strong negative correlation
Malicious_Votes ↔ Reputation	−0.823	Strong negative correlation
Domain_Length ↔ Threat_Severity	+0.134	WEAK positive correlation
Has_Hyphen ↔ Threat_Severity	+0.089	NEGLIGIBLE correlation
Has_Numbers ↔ Threat_Severity	+0.042	NEGLIGIBLE correlation

§4 Engineered Features

The following ratio-based normalized metrics were engineered from raw voting counts:

- Malicious_Ratio = Malicious_Votes / Total_Engines (Mean: 0.0144)
- Harmless_Ratio = Harmless_Votes / Total_Engines (Mean: 0.4274)
- Detection_Confidence = (Malicious_Votes + Suspicious_Votes) / Total_Engines (Mean: 0.0162)
- Trust_Ratio = Harmless_Votes / Total_Engines

§5 Regression Models — Five Models Evaluated

Model	RMSE	R ² Score	Overfitting Risk	Practical Utility
Gradient Boosting	0.0000	1.0000	EXTREME	No — Overfitting
Random Forest (BEST)	0.1410	0.8390	Low	YES — Recommended
Lasso (α=0.01)	0.3080	0.2570	Low	Limited
Ridge (α=1.0)	0.3210	0.1900	Low	Limited
Linear Regression	0.3350	0.1310	Low	Limited

§6 Model Evaluation

6.1 Random Forest Performance (Recommended Model)

- Training R²: 0.844 Test R²: 0.839 Train-Test Gap: 0.5% (excellent) - RMSE: 0.141 Mean Residual: −0.0089 (unbiased) 5-Fold CV Mean R²: 0.836 (σ = 0.019) — robust generalization - Cross-Validation Fold Results: 0.823, 0.847, 0.836, 0.841, 0.835

6.2 Feature Importance (Random Forest)

Rank	Feature	Importance	Percentage	Interpretation
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1	Malicious_Votes	0.4832	48.3%	DOMINANT predictor
2	Malicious_Ratio	0.1847	18.5%	Normalized malicious signal
3	Undetected_Votes	0.0934	9.3%	Engines without clear verdict
4	Detection_Confidence	0.0687	6.9%	Combined malicious+suspicious
5	Reputation	0.0512	5.1%	Reputation metric utility limited
6	Domain_Length	0.0234	2.3%	Domain structure weak signal
7	Has_Numbers	0.0108	1.1%	Character composition negligible
8	Has_Hyphen	0.0045	0.5%	Negligible predictive value

§7 Discussion & Conclusion

7.1 Crowd-Sourced Voting Superiority

Malicious_Votes alone (48.3% importance) comprises nearly half of all predictive signal. The top 2 features (Malicious_Votes + Malicious_Ratio) account for 66.8% of model decisions. Domain structural features provide minimal utility (<2% importance) despite common security folklore.

7.2 Gradient Boosting Overfitting

The Gradient Boosting model's perfect $R^2=1.0$ indicates memorization rather than generalization on the small 162-record dataset. With 200 boosting iterations, the model fitted exact decision boundaries for every training sample. Model complexity must scale with dataset size.

7.3 Strategic Recommendations

- Deploy Random Forest Regressor ($R^2=0.839$, $RMSE=0.141$) for operational domain triage
- Implement SMOTE to balance training data for improved minority class performance
- Integrate model predictions into SOC alert prioritization workflows
- Expand dataset beyond 162 domains to improve statistical power
- Incorporate temporal features: domain age, registration date anomalies, DNS TTL patterns